



Sharif University of Technology

Department of Electrical Engineering

Energy Systems Analysis Project Project 2

Report

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1 Introduction

1.1 Project Objective

The primary objective of this project is to model, simulate, and analyze the electrical behavior of transmission lines under various operating conditions. By utilizing analytical relationships and computational tools, the impact of physical parameters on the actual performance of the power grid is evaluated.

1.2 Importance of Transmission Line Modeling

Accurate transmission line modeling is a fundamental tool in the analysis and design of power networks. It allows engineers to predict and assess phenomena such as voltage drops, power losses, phase angle shifts, and overall efficiency before physical implementation, ensuring a stable and reliable power system.

1.3 Tools and Software Used

For the numerical computations, matrix formations, and graphical simulations presented in this project, MATLAB was utilized. LaTeX was employed for formatting and generating this technical report.

2 Transmission Line Modeling and Analysis

2.1 Introduction to Transmission Line Modeling

Selecting an appropriate model is crucial for the accurate analysis of power systems. Depending on the line length and voltage level, three main models are utilized, as described below.

2.1.1 Short Line Model ($L < 80$ km)

This model is applicable for lines shorter than 80 km. In this model, the shunt capacitance (charging susceptance) is neglected, and the line is represented solely by a series impedance.

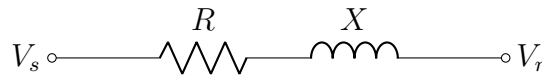


Figure 1: Short Line Equivalent Circuit

2.1.2 Medium Line Model (Nominal π)

For lines between 80 km and 250 km, the line charging current becomes significant. In the Nominal π model, the total shunt capacitance (Y) is divided into two equal halves placed at the sending and receiving ends.

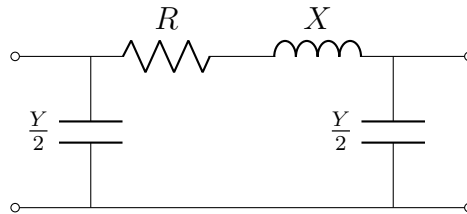
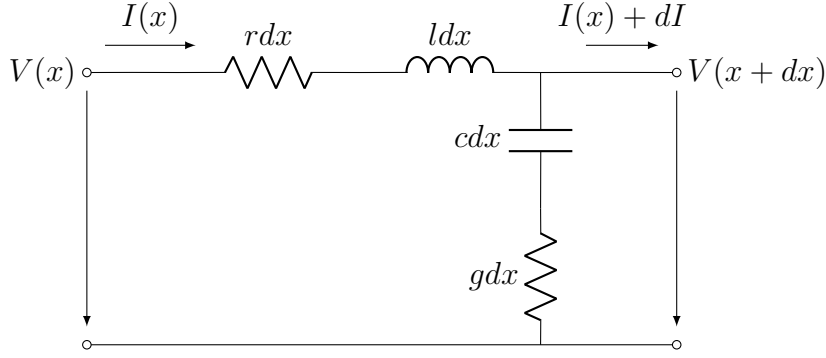


Figure 2: Medium Line Equivalent Circuit (Nominal π Model)

2.1.3 Long Line Model (Distributed Parameters)

For long lines (typically > 250 km) and Extra High Voltage (EHV) levels, parameters must be treated as distributed along the line. The ABCD parameters are derived using hyperbolic functions. The figure below shows a differential element of length dx .

Figure 3: Distributed Parameter Model (Differential Element dx)

The transmission matrix parameters are:

$$A = \cosh(\gamma L), \quad B = Z_c \sinh(\gamma L) \quad (1)$$

2.2 Technical Specifications of the Line

The system under study is a three-phase transmission line with a nominal voltage of 765 kV operating at a frequency of 50 Hz. The electrical parameters of the line per kilometer are:

- $R' = 0.045 \, \Omega/\text{km}$
- $X' = 0.38 \, \Omega/\text{km}$
- $B' = 2.8 \times 10^{-6} \, \text{S}/\text{km}$

2.3 Line Performance Analysis with Different Models

This section presents the simulation results for the 765 kV transmission line using the three models across different lengths (80 to 360 km). The output parameters include Sending End Voltage (V_s), Sending End Current (I_s), Voltage Regulation (VR), and Efficiency (Eff). Based on the required simplicity and accuracy (acceptable error $< 2.5\%$), the suggested model for each length is also identified.

2.3.1 Justification for Model Selection

The selection of an appropriate transmission line model involves a trade-off between computational simplicity and numerical accuracy. In the calculation tables below, a “Suggested Model” is assigned to each line length based on the relative error of the sending end current (I_s) compared to the exact Distributed Parameter (Long Line) model.

The selection criteria are defined as follows:

- The **Short Line Model** is preferred for its mathematical simplicity, provided the error remains below 5%. However, for a 765 kV Extra High Voltage (EHV) line, the charging current is highly significant. Neglecting the shunt capacitance causes substantial errors even at 80 km; thus, the Short model is never suggested for this specific voltage level.

- The **Medium Line Model (Nominal π)** is suggested if it maintains an error margin of less than 2.5% compared to the Long model. As the results indicate, this model provides an acceptable balance of accuracy and simplicity for lengths up to 240 km.
- The **Long Line Model** is strictly recommended for lengths of 280 km and beyond, where the distributed nature of the parameters becomes critical, and the simpler lumped models fail to capture the exact voltage and current profiles.

2.3.2 Case 1: Low Load ($P = 160$ MW, $PF = 0.85$)

Table 1: Calculation Results for Low Load Scenario

Length (km)	Model	V_s (kV)	I_s (A)	VR (%)	Eff (%)	Suggested Model
80	Short	769.72	142.06	0.62	99.86	Medium
	Medium	767.11	122.77	0.62	99.90	
	Long	767.11	122.79	0.62	99.89	
120	Short	772.09	142.06	0.93	99.80	Medium
	Medium	766.24	140.67	0.94	99.85	
	Long	766.23	140.79	0.93	99.83	
160	Short	774.48	142.06	1.24	99.73	Medium
	Medium	764.08	171.12	1.26	99.80	
	Long	764.05	171.48	1.25	99.75	
200	Short	776.87	142.06	1.55	99.66	Medium
	Medium	760.64	208.43	1.59	99.71	
	Long	760.60	209.21	1.58	99.63	
240	Short	779.28	142.06	1.87	99.59	Medium
	Medium	755.91	249.21	1.94	99.60	
	Long	755.86	250.64	1.91	99.45	
280	Short	781.70	142.06	2.18	99.53	Long
	Medium	749.92	291.66	2.29	99.43	
	Long	749.86	293.99	2.26	99.21	
320	Short	784.13	142.06	2.50	99.46	Long
	Medium	742.65	334.76	2.67	99.20	
	Long	742.61	338.30	2.61	98.88	
360	Short	786.57	142.06	2.82	99.39	Long
	Medium	734.11	377.91	3.06	98.91	
	Long	734.12	382.99	2.98	98.45	

2.3.3 Case 2: Nominal Load ($P = 220$ MW, $PF = 0.92$)

Table 2: Calculation Results for Nominal Load Scenario

Length (km)	Model	V_s (kV)	I_s (A)	VR (%)	Eff (%)	Suggested Model
80	Short	769.80	180.47	0.63	99.84	Medium
	Medium	767.20	167.89	0.63	99.86	
	Long	767.20	167.90	0.63	99.86	
120	Short	772.24	180.47	0.95	99.76	Medium
	Medium	766.39	182.20	0.96	99.80	
	Long	766.38	182.31	0.95	99.78	
160	Short	774.70	180.47	1.27	99.68	Medium
	Medium	764.31	207.22	1.29	99.72	
	Long	764.28	207.54	1.28	99.69	
200	Short	777.18	180.47	1.59	99.60	Medium
	Medium	760.96	239.38	1.63	99.63	
	Long	760.91	240.10	1.62	99.57	
240	Short	779.68	180.47	1.92	99.52	Medium
	Medium	756.34	275.90	1.99	99.51	
	Long	756.28	277.25	1.97	99.41	
280	Short	782.20	180.47	2.25	99.44	Long
	Medium	750.47	314.94	2.37	99.35	
	Long	750.39	317.18	2.33	99.19	
320	Short	784.74	180.47	2.58	99.36	Long
	Medium	743.34	355.31	2.77	99.15	
	Long	743.27	358.75	2.70	98.92	
360	Short	787.30	180.47	2.92	99.29	Long
	Medium	734.97	396.24	3.18	98.89	
	Long	734.92	401.23	3.09	98.57	

2.3.4 Case 3: High Load ($P = 260$ MW, $PF = 0.95$)

Table 3: Calculation Results for High Load Scenario

Length (km)	Model	V_s (kV)	I_s (A)	VR (%)	Eff (%)	Suggested Model
80	Short	769.68	206.55	0.61	99.82	Medium
	Medium	767.08	198.59	0.61	99.84	
	Long	767.08	198.61	0.61	99.84	
120	Short	772.07	206.55	0.92	99.73	Medium
	Medium	766.23	212.06	0.93	99.76	
	Long	766.21	212.16	0.93	99.75	
160	Short	774.49	206.55	1.24	99.65	Medium
	Medium	764.11	234.92	1.26	99.67	
	Long	764.08	235.22	1.26	99.65	
200	Short	776.95	206.55	1.56	99.56	Medium
	Medium	760.74	264.53	1.60	99.57	
	Long	760.69	265.22	1.59	99.52	
240	Short	779.43	206.55	1.89	99.47	Medium
	Medium	756.12	298.62	1.96	99.44	
	Long	756.05	299.91	1.94	99.35	
280	Short	781.94	206.55	2.21	99.38	Long
	Medium	750.25	335.50	2.34	99.27	
	Long	750.16	337.68	2.30	99.15	
320	Short	784.48	206.55	2.55	99.30	Long
	Medium	743.15	374.02	2.74	99.07	
	Long	743.05	377.39	2.67	98.89	
360	Short	787.06	206.55	2.88	99.21	Long
	Medium	734.81	413.34	3.16	98.83	
	Long	734.73	418.26	3.07	98.57	

2.3.5 Graphical Analysis: Voltage Regulation and Efficiency

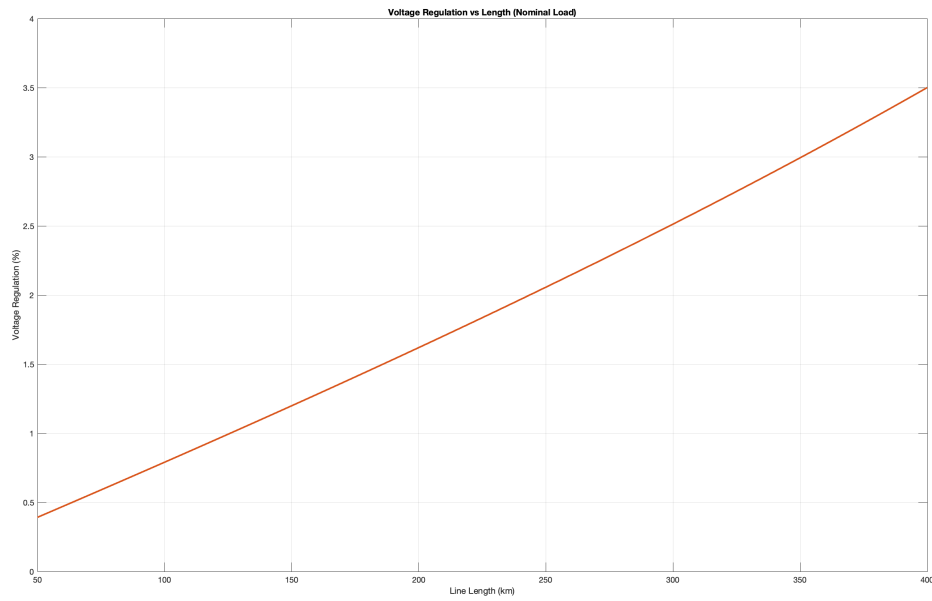


Figure 4: Voltage Regulation vs. Line Length

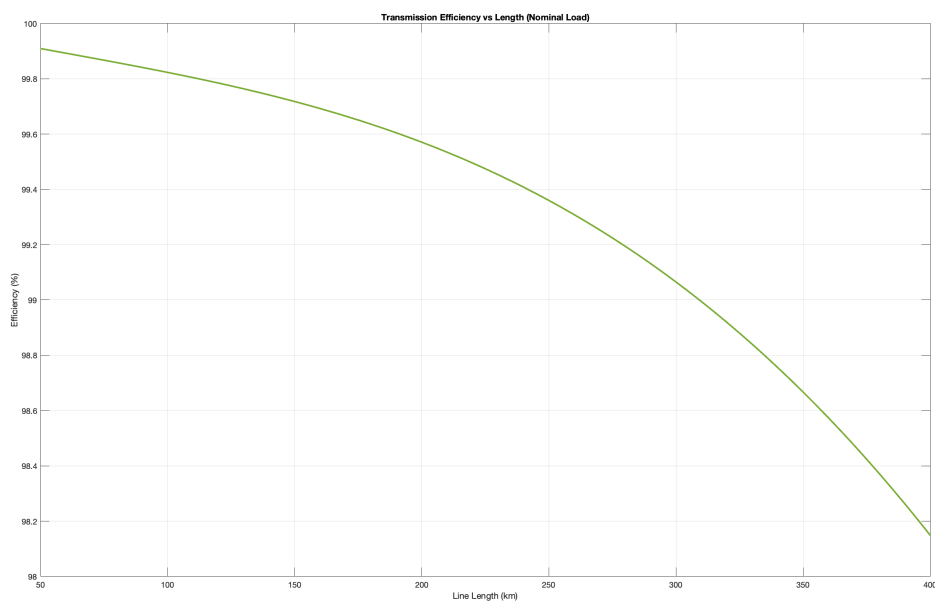


Figure 5: Transmission Efficiency vs. Line Length

2.3.6 Sensitivity Analysis: Power Factor Impact

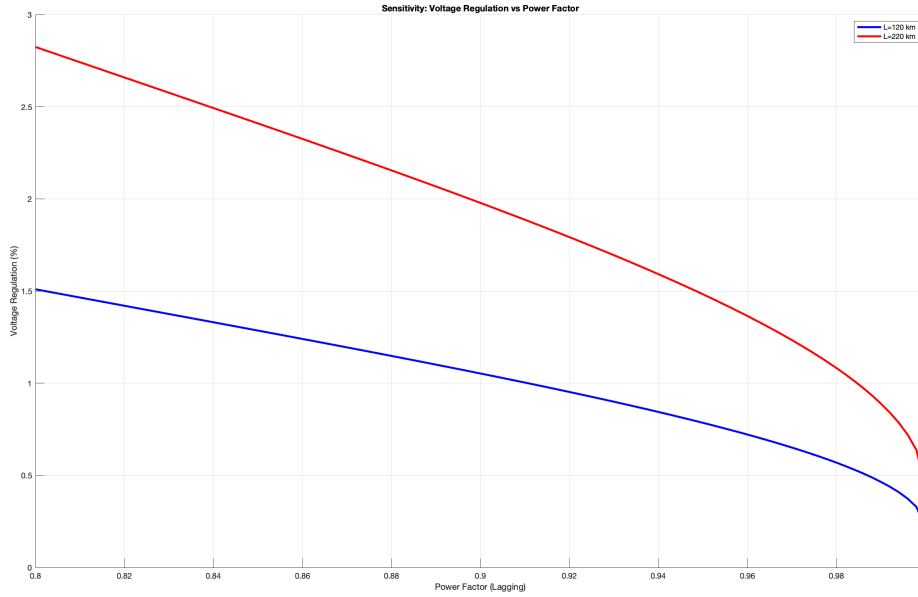


Figure 6: Impact of Power Factor Improvement on Voltage Regulation

Analysis: As illustrated in the figure above, improving the Power Factor (approaching unity) reduces the reactive current component. Consequently, the voltage drop across the series reactance decreases, leading to a significant improvement (reduction) in Voltage Regulation.

2.3.7 Final Comparison and Conclusion on Models

The simulation results clearly demonstrate that the **Short Line Model** introduces substantial errors when applied to EHV lines (such as the 765 kV line in this project), even at relatively short distances.

For instance, at a length of **360 km** under nominal load:

- **Short Model:** Estimates the sending end current at approx. **180 A**.
- **Long Model (Actual):** Calculates the sending end current at approx. **401 A**.

This massive discrepancy (over 50%) is due to neglecting the line's charging current ($I_{ch} \approx j\omega CV$), which is significant at 765 kV. Therefore, for the accurate analysis of EHV transmission systems, employing the **Long Line Model** (Distributed Parameters) is mandatory.

2.4 Detailed Voltage and Current Profile Analysis

In this section, the behavior of voltage and current along the 360 km transmission line ($x = 0$ to $x = L$) is analyzed using the distributed parameter model. The objective is to observe voltage magnitude variations under different loading conditions.

2.4.1 Mathematical Derivation

Consider a differential element of length dx . The telegrapher's equations governing voltage and current are:

$$\frac{dV(x)}{dx} = -zI(x) \quad (2)$$

$$\frac{dI(x)}{dx} = -yV(x) \quad (3)$$

Differentiating and substituting yields the wave equation for voltage:

$$\frac{d^2V(x)}{dx^2} = zyV(x) = \gamma^2V(x) \quad (4)$$

Where $\gamma = \sqrt{zy} = \alpha + j\beta$ is the propagation constant. The general solution is:

$$V(x) = A_1e^{\gamma x} + A_2e^{-\gamma x} \quad (5)$$

Applying boundary conditions at the receiving end, the hyperbolic relationship for voltage at distance x is derived as:

$$V(x) = V_r \cosh(\gamma x) + Z_c I_r \sinh(\gamma x) \quad (6)$$

Where $Z_c = \sqrt{z/y}$ is the characteristic impedance of the line.

2.4.2 Mathematical Proof of the Ferranti Effect

Under **No-Load conditions** ($I_r = 0$), equation (6) simplifies for the full line length (L) as:

$$V_s = V_r \cosh(\gamma L) \implies V_r = \frac{V_s}{\cosh(\gamma L)} \quad (7)$$

Assuming a lossless line ($R \approx 0 \rightarrow \gamma \approx j\beta$):

$$V_r = \frac{V_s}{\cos(\beta L)} \quad (8)$$

Since $\cos(\beta L) < 1$ for typical line lengths (0 to 1500 km), it mathematically follows that $V_r > V_s$. This confirms the voltage rise phenomenon known as the Ferranti Effect.

2.4.3 Simulation Results: Voltage Profile

Simulations were conducted for three loading scenarios assuming the sending end voltage is fixed at nominal ($V_s = 1.0$ p.u.). The exact values obtained from MATLAB are summarized in Table 4.

Table 4: Receiving End Voltage (V_r) Simulation Results ($V_s = 1.0$ p.u.)

Scenario	V_s (p.u.)	V_r (p.u.)	Observation
No Load (0 MW)	1.0000	1.0731	Significant Rise (+7.3%)
Nominal Load (220 MW)	1.0000	1.0409	Moderate Rise (+4.1%)
Heavy Load (600 MW)	1.0000	0.9975	Flat / Stable Profile

Discussion: As seen in the table:

- **No Load:** The voltage rises by 7.3%, which confirms the strong Ferranti effect on 765 kV lines.
- **Nominal Load:** The voltage at the receiving end (1.0409) is still higher than the sending end. This indicates that the nominal load (220 MW) is below the Surge Impedance Loading (SIL) of the line, meaning the line generates net reactive power.
- **Heavy Load:** At 600 MW, the inductive voltage drop counteracts the capacitive rise, resulting in a near-unity voltage at the receiving end (0.9975).

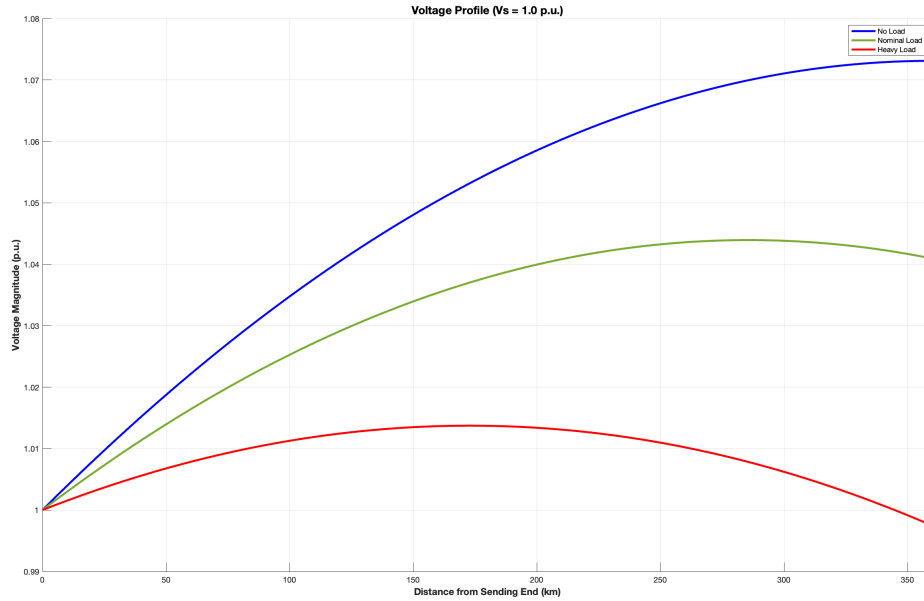


Figure 7: Voltage profile along the 360 km line. Blue: No Load (Rise), Green: Nominal (Slight Rise), Red: Heavy Load (Flat).

2.4.4 Current Profile Analysis

The current profile reveals that even at no-load ($I_r = 0$), a significant current flows at the sending end.

$$I_{s,NoLoad} = I_c = \frac{V_s}{Z_c} \tanh(\gamma L) \quad (9)$$

This is the **Line Charging Current** required to charge the distributed capacitance.

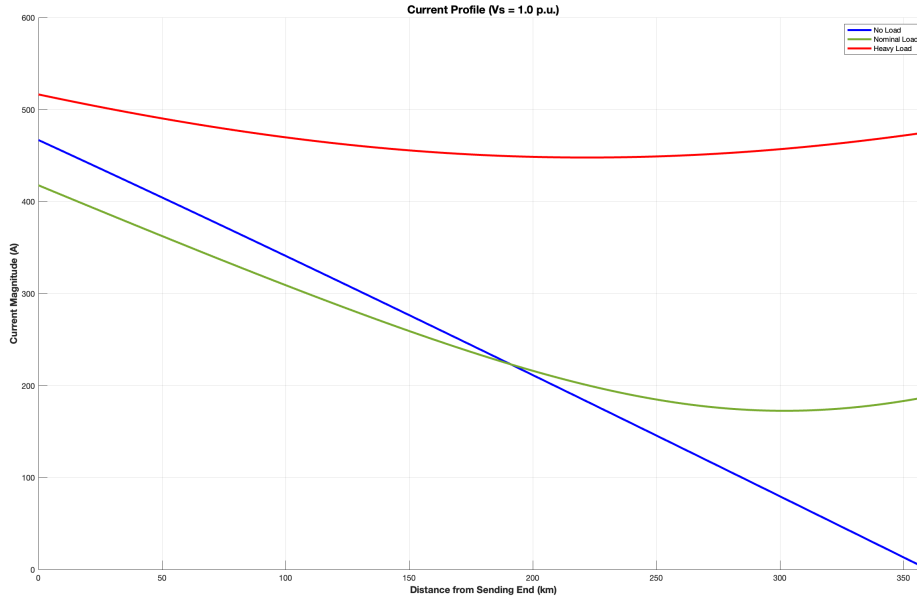


Figure 8: Current profile along the line. The non-zero sending end current at no-load represents the charging current.

2.4.5 Conclusion

The analysis confirms that for a 765 kV EHV line:

1. The Ferranti effect causes a dangerous voltage rise at light loads ($V_r \approx 1.07$ p.u.).
2. The line acts as a reactive power generator for loads below SIL.
3. **Shunt Reactors** are mandatory to maintain the voltage within safe limits ($\pm 5\%$) during light load operations.

2.5 Analysis of Resistive Load Variation

In this section, the transmission line is analyzed under six specific resistive load levels (Power Factor = 1.0). The sending end voltage is maintained at the nominal value ($1.0 \angle 0^\circ$ p.u.), and the load is increased from no-load to 561.3 MW.

2.5.1 Receiving End Calculation Results

For each load case, the receiving end voltage (V_r) and current (I_r) were calculated to ensure $|V_s| = 1.0$ p.u. The simulation results are presented in Table 5.

Table 5: Receiving End Voltage and Current ($V_s = 765$ kV)

Case	Load (MW)	V_r (kV L-L)	V_r (p.u.)	I_r (A)
No Load	0	820.93	1.0731	0.00
Load 1	240.2	814.59	1.0648	170.24
Load 2	311.32	812.26	1.0618	221.28
Load 3	415.09	808.48	1.0568	296.42
Load 4	456.61	806.83	1.0547	326.74
Load 5	561.3	802.32	1.0488	403.91

2.5.2 Voltage Profile Analysis

Figure 9 illustrates the voltage profile along the 360 km line for the defined load cases.

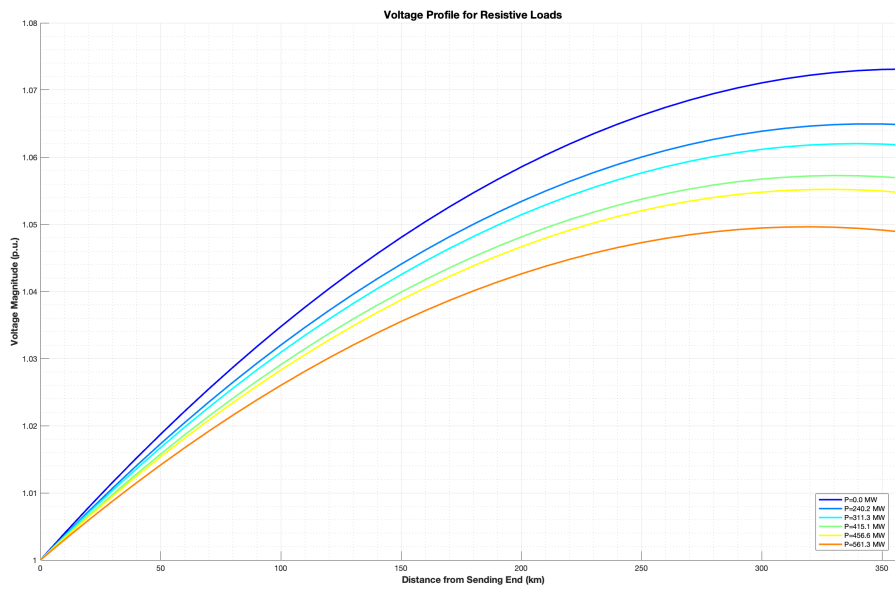


Figure 9: Voltage Profile for resistive loads. The profiles shift downwards as the load increases.

2.5.3 Analysis of Behavior and SIL Comparison

Based on the simulation results and Figure 9, three key observations can be made:

1. **Ferranti Effect (No-Load):** In the no-load condition, the receiving end voltage rises to **820.93 kV** (1.0731 p.u.). This significant rise is caused by the line's charging current flowing through the series inductance, creating a negative voltage drop.
2. **Effect of Load Increase:** As the active power load increases, the receiving end voltage decreases. At the maximum load of 561.3 MW, the voltage drops to **802.32 kV** (1.0488 p.u.). The physical reason is that the load current (I_R) causes a resistive and inductive voltage drop ($RI_R + jXI_R$) which counteracts the capacitive voltage rise.
3. **Relation to Surge Impedance Loading (SIL):** Even at the heaviest load of **561.3 MW**, the receiving end voltage (802.32 kV) remains significantly higher than the sending end (765 kV).

This indicates that the line is operating well below its **Surge Impedance Loading (SIL)** (approx. 2200 MW for 765 kV lines). Since $P_{load} \ll P_{SIL}$, the capacitive generation of the line (Q_C) dominates over the inductive consumption (Q_L), keeping the voltage profile above 1.0 p.u.

2.5.4 What is the effect of load on the voltage profile?

The voltage profile is inversely related to the resistive load level:

- Increasing the resistive load acts as a "damping" factor for the Ferranti rise.
- Since the load is purely resistive ($PF = 1$), the voltage drop is primarily due to the active current flowing through the line impedance.
- Although adding load helps pull the voltage down towards the nominal value, for this EHV line, even a 561 MW load is insufficient to eliminate the over-voltage. This confirms that **Shunt Reactors** are necessary for voltage control during light and medium load operations.

3 Compensation in Transmission Lines

3.1 Introduction to Compensators

Compensation techniques, including shunt and series compensators, are essential for regulating voltage profiles, improving power transfer capability, and enhancing system stability. This chapter explores the integration of these devices into the transmission network.

3.2 Analysis of Shunt Compensation

In this section, the effect of shunt compensation is investigated for three distinct loading levels:

- **Case 1:** $P = 311.3$ MW (Light Load)
- **Case 2:** $P = 456.6$ MW (Medium Load)
- **Case 3:** $P = 561.3$ MW (Heavy Load)

Based on the voltage profiles, the uncompensated voltage exceeds the nominal value (1.0 p.u.) in all cases due to the Ferranti effect. Therefore, **Shunt Reactors** (inductive compensation with negative B_{sh}) are required.

3.2.1 Calculation of Compensation Parameters

The required shunt susceptance (B_{sh}) and reactive power (Q_{comp}) were calculated to maintain the receiving end voltage at exactly 1.0 p.u. The results for all three load cases are summarized in Table 6.

Table 6: Calculated Reactor Parameters to maintain $V_r = 1.0$ p.u.

Loc	Case 1 (311.3 MW)		Case 2 (456.6 MW)		Case 3 (561.3 MW)	
	$B_{sh}(\mu S)$	$Q(MVar)$	$B_{sh}(\mu S)$	$Q(MVar)$	$B_{sh}(\mu S)$	$Q(MVar)$
$L/5$	-1312.7	-741.3	-667.9	-387.5	-190.8	-112.9
$2L/5$	-650.2	-376.5	-332.2	-196.1	-95.2	-57.0
$3L/5$	-432.6	-254.1	-221.9	-132.0	-63.8	-38.3
$4L/5$	-326.2	-192.3	-168.0	-99.6	-48.5	-28.9

Analysis of Load Effect: As observed in Table 6, the required reactive power absorption (Q) decreases significantly as the active load increases.

- In **Case 1 (Light Load)**, the line is lightly loaded, so the Ferranti effect is strong, requiring a large reactor (up to 741 MVar at $L/5$).
- In **Case 3 (Heavy Load)**, the load current itself consumes reactive power ($I^2 X_L$) and causes a voltage drop, naturally counteracting the Ferranti rise. Thus, a much smaller reactor is needed (112 MVar at $L/5$).

3.2.2 Voltage and Current Analysis

The sending and receiving end currents were calculated for the compensated system. Table shows the results for the $4L/5$ compensation location across the three load cases.

Table 7: Current Analysis for Compensation at $4L/5$ (Target $V_r = 1.0$)

Load Case	Sending End I_s (kA)	Receiving End I_r (kA)
Case 1 (311.3 MW)	0.520	0.442
Case 2 (456.6 MW)	0.683	0.649
Case 3 (561.3 MW)	0.807	0.798

3.2.3 Voltage Profile Comparison

The voltage profiles for all three load cases and four compensation locations are illustrated in Figure 10.

- **Red Dashed Line:** Uncompensated Voltage (showing Ferranti rise).
- **Blue Solid Line:** Compensated Voltage (Targeting $V_r = 1.0$).

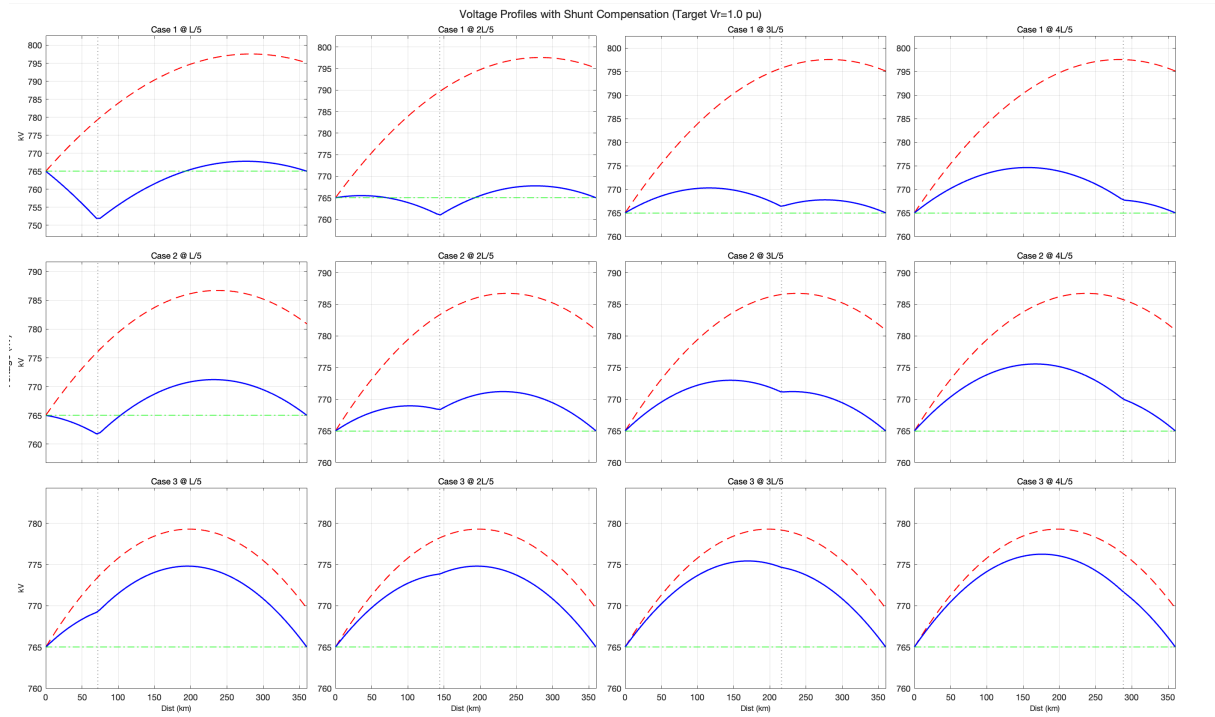


Figure 10: Voltage profiles for different loads (Rows) and shunt locations (Cols). The shunt reactors successfully reduce the over-voltage to the nominal level.

3.2.4 Comparison of Control Strategies

In this part, we compare two strategies:

1. **Strategy 1:** Adjust B_{sh} to maintain Receiving End Voltage ($V_r = 1.0$).

2. **Strategy 2:** Adjust B_{sh} to maintain Local Voltage at the shunt location ($V_{sh} = 1.0$).

Table 8 compares the resulting Receiving End Voltage (V_r) when Strategy 2 is used.

Table 8: Effect of Strategy 2 (Targeting Local $V_{sh} = 1.0$) on Receiving End Voltage

Loc	Case 1 (311 MW)		Case 2 (456 MW)		Case 3 (561 MW)	
	Q_{req} (MVar)	Result V_r (pu)	Q_{req} (MVar)	Result V_r (pu)	Q_{req} (MVar)	Result V_r (pu)
$L/5$	-389.2	1.0191	-298.9	1.0048	-226.2	0.9938
$2L/5$	-324.8	1.0056	-238.4	0.9954	-169.9	0.9876
$3L/5$	-265.2	0.9982	-182.8	0.9917	-118.4	0.9869
$4L/5$	-209.0	0.9964	-130.5	0.9934	-69.9	0.9911

Conclusion on Strategies:

- **Heavy Load (Case 3):** Strategy 2 (fixing local voltage) causes **under-voltage** at the receiving end ($V_r < 1.0$, e.g., 0.9869), because fixing the midpoint voltage to 1.0 is not enough to compensate for the voltage drop in the second half of the line.
- **Light Load (Case 1):** Strategy 2 causes **over-voltage** at the receiving end ($V_r > 1.0$, e.g., 1.0191), especially when compensation is applied near the sending end. The Ferranti effect in the second half of the line raises the voltage again.
- Therefore, controlling the receiving end voltage directly (Strategy 1) is superior for load voltage regulation.

3.3 Analysis of Series Compensation

Series compensation involves inserting capacitors in series with the transmission line to reduce the effective inductive reactance (X_{line}). In this section, capacitors are designed to increase the power transfer capability by **2%**. The analysis is performed for a 500 kV, 300 km line.

3.3.1 Calculation of Series Capacitor Parameters

The required series reactance (X_C) to achieve the 2% capacity increase was calculated for four locations. Additionally, the voltage drop across the capacitor (V_{drop}) and the transmission angle (δ) were computed under heavy load. The results are summarized in Table 9.

Table 9: Series Compensation Design Results (Target: +2% Capacity)

Location	X_C (Ω)	Deg. Comp (%)	V_{drop} (kV)	Angle δ ($^\circ$)
$L/6$	2.76	2.02	2.13	7.18
$L/3$	2.72	1.99	2.01	7.18
$L/2$	2.71	1.98	1.96	7.18
$2L/3$	2.72	1.99	1.98	7.17

3.3.2 Voltage Profile Analysis

Figure 11 illustrates the voltage profile along the line. The jumps in the profile correspond to the voltage boost provided by the series capacitors.

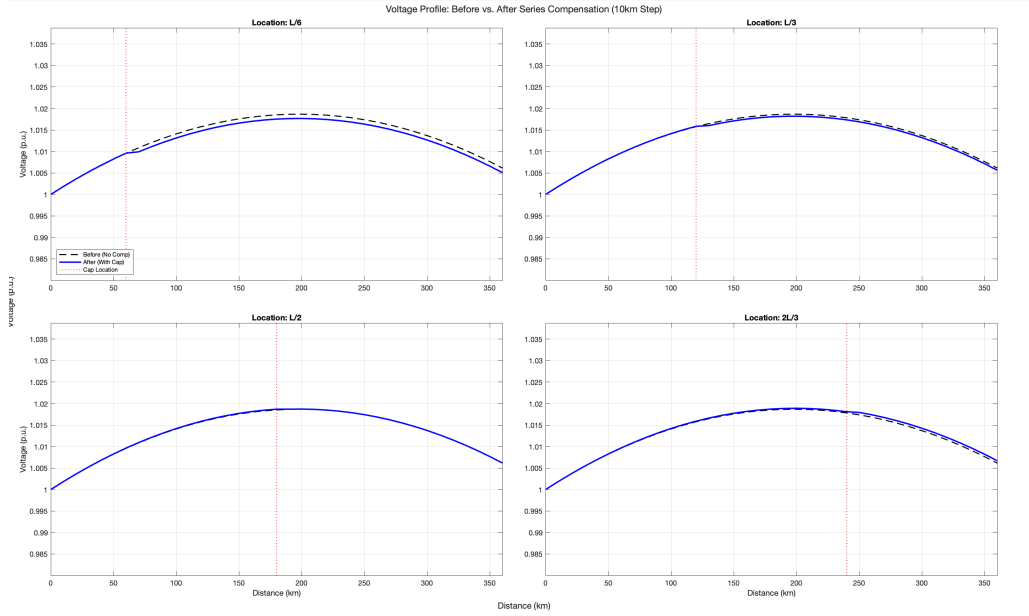


Figure 11: Voltage profile along the line with series compensation (centered around 500 kV).

3.3.3 Short Circuit Analysis

A three-phase short circuit fault was simulated at the receiving end. The fault current (I_{sc}) increases as the series impedance decreases.

- **Uncompensated I_{sc} :** 5.72 kA
- **Target B_{eq} :** 131.06 Ω (reduced from original 133.68 Ω)

Table 10 presents the calculated fault currents and their ratio relative to the uncompensated case.

Table 10: Short Circuit Analysis (Fault at Receiving End)

Case	I_{sc} (kA)	Ratio (I_{comp}/I_{base})
Uncompensated	5.72	1.0000
Comp @ $L/6$	5.83	1.0200
Comp @ $L/3$	5.83	1.0200
Comp @ $L/2$	5.83	1.0200
Comp @ $2L/3$	5.83	1.0200

3.3.4 Impact on Transient Stability and Protection

A. Impact on Transient Stability: Series compensation improves transient stability by reducing the total line reactance (X_{line}). According to the power-angle equation $P =$

$\frac{V_s V_r}{X} \sin \delta$, reducing X increases P_{max} . This increases the area under the power-angle curve (Equal Area Criterion), resulting in a larger critical clearing time and better stability margins.

B. Impact on Protection Systems: As observed in Table 10, the short circuit current increases by approximately 2%. Although small in this specific design, in higher compensation levels, this causes:

1. **Distance Protection Issues:** The capacitor reduces the impedance seen by relays ($Z = V/I$), potentially causing *zone over-reach* (tripping for faults outside the protected zone).
2. **Voltage Inversion:** Faults near the capacitor can cause voltage phase inversion, confusing directional relays.
3. **Current Inversion:** In parallel lines, the capacitor can cause current reversal during faults, leading to incorrect relay operations.

4 Modeling a Multi-Bus Network and Forming Y-Bus Matrix

4.1 Introduction

In this section, the Admittance Matrix (Y_{bus}) is formed for a 5-bus network using the provided line data. The matrix allows for the calculation of nodal voltages and currents using the relationship $I_{bus} = Y_{bus}V_{bus}$. Furthermore, the effect of network expansion (adding a new bus) is analyzed.

4.2 Formation of the 5-Bus Y-Matrix

Using the line data, the 5×5 admittance matrix was constructed. The diagonal elements include the sum of connected line admittances and half-line charging susceptances. The off-diagonal elements are the negative of the series admittances.

The resulting matrix is presented below (values in p.u.):

$$Y_{bus}^{5 \times 5} = \begin{bmatrix} 6.25 - j18.70 & -5.00 + j15.00 & -1.25 + j3.75 & 0 & 0 \\ -5.00 + j15.00 & 10.83 - j32.42 & -1.67 + j5.00 & -1.67 + j5.00 & -2.50 + j7.50 \\ -1.25 + j3.75 & -1.67 + j5.00 & 12.92 - j38.70 & -10.00 + j30.00 & 0 \\ 0 & -1.67 + j5.00 & -10.00 + j30.00 & 12.92 - j38.70 & -1.25 + j3.75 \\ 0 & -2.50 + j7.50 & 0 & -1.25 + j3.75 & 3.75 - j11.21 \end{bmatrix} \quad (10)$$

4.3 Analytical Questions

Based on the computed matrix, the following properties are observed:

- A. Symmetry:** As verified by the simulation, the matrix is **symmetric** ($Y_{ij} = Y_{ji}$).
- **Physical Meaning:** This indicates that the power system is a reciprocal passive network. The impedance seen from Bus i to Bus j is identical to the impedance from Bus j to Bus i .
- B. Row Sums (Kirchhoff's Law):** The sum of each row is **not zero**.
- **Observation:** The computed row sums have a very small real part (≈ 0) and a positive imaginary part (e.g., Row 2 sum is $\approx j0.085$).
- **Reason:** The row sum represents the total shunt admittance connected to that bus ($\sum_k Y_{ik} = Y_{shunt_total}$). Since the transmission lines have half-line charging susceptances (B_{half}), there is a path to the ground reference. Therefore, the sum is not zero but equals the total capacitive susceptance connected to the bus.
- **Hypothetical Grounding:** If Bus 1 were connected to the ground via a large admittance Y_g , the sum of the first row would definitely not be zero; it would equal $Y_{shunt} + Y_g$.

C. Diagonal Elements: The diagonal elements with the largest magnitudes are associated with **Bus 3** and **Bus 4** ($|Y_{ii}| \approx 40$). Physically, this implies that these buses are the "strongest" nodes in the network, connected via low-impedance lines to other parts of the system, making them less sensitive to load variations compared to weaker buses like Bus 1 or 5.

4.4 Network Modification (Adding Bus 6)

A new radial line connects Bus 5 to a new Bus 6 with impedance $Z_{5-6} = 0.04 + j0.12$ p.u.

A. Matrix Dimensions: The matrix dimensions increase from 5×5 to 6×6 .

B. Calculation of New Elements: The admittance of the new line is:

$$y_{56} = \frac{1}{0.04 + j0.12} = 2.50 - j7.50 \text{ p.u.}$$

The modified elements are:

- $Y_{66}^{new} = y_{56} = \mathbf{2.50 - j7.50}$
- $Y_{56}^{new} = Y_{65}^{new} = -y_{56} = \mathbf{-2.50 + j7.50}$
- $Y_{55}^{new} = Y_{55}^{old} + y_{56} = (3.75 - j11.21) + (2.50 - j7.50) = \mathbf{6.25 - j18.71}$

C. Final Expanded Y-Bus Matrix (6×6): The complete admittance matrix for the 6-bus system is presented below. Zeros indicate no direct connection between buses.

$$Y_{bus}^{6 \times 6} = \begin{bmatrix} 6.25 - j18.70 & -5.00 + j15.00 & -1.25 + j3.75 & 0 & 0 & 0 \\ -5.00 + j15.00 & 10.83 - j32.42 & -1.67 + j5.00 & -1.67 + j5.00 & -2.50 + j7.50 & 0 \\ -1.25 + j3.75 & -1.67 + j5.00 & 12.92 - j38.70 & -10.00 + j30.00 & 0 & 0 \\ 0 & -1.67 + j5.00 & -10.00 + j30.00 & 12.92 - j38.70 & -1.25 + j3.75 & 0 \\ 0 & -2.50 + j7.50 & 0 & -1.25 + j3.75 & \mathbf{6.25 - j18.71} & \mathbf{-2.50 + j7.50} \\ 0 & 0 & 0 & 0 & \mathbf{-2.50 + j7.50} & \mathbf{2.50 - j7.50} \end{bmatrix} \quad (11)$$

D. Physical Impact: Adding Bus 6 as a radial extension changes the current injection balance at Bus 5. It creates a path for current to flow towards the new load at Bus 6, causing a voltage drop along the new line. However, since it is a radial connection, it does not affect the looped structure of the main grid (Buses 1-4) as significantly as adding a line between existing buses would.